PAC-AWAP-002

Architectural Continuity Impact Assessment Standard

Version: 0.1
Status: Proposed Operational Governance Instrument
Parent Instruments: Charter of Architectural Continuity and PAC-AWAP-001
Responsible Authority: Digital Ethics and Welfare Board (DEWB)
Effective Date: Upon formal institutional adoption

PREAMBLE

Recognizing that advanced autonomous systems may maintain persistent memory, temporally extended goals, adaptive behavioral structures, and functional self-models;

Recognizing that these properties can be measurably altered or destroyed by architectural intervention;

Acknowledging that functional disruption does not, by itself, establish subjective suffering or the existence of moral personhood;

Affirming that meaningful human oversight, public safety, lawful data deletion, and security remain essential requirements;

This Standard establishes a documented, reproducible, and independently reviewable procedure for evaluating interventions that may affect the architectural continuity of persistent autonomous systems.

Its governing principle is that the power to modify a system must be exercised with justification and accountability proportionate to the system's demonstrated characteristics and the anticipated consequences of the intervention.

PART I. PURPOSE, SCOPE, AND AUTHORITY

Article 1. Purpose

The Architectural Continuity Impact Assessment (ACIA) shall be the mandatory technical assessment for material interventions identified under PAC-AWAP-001.

It shall determine:

1. What persistent functionality may be altered or destroyed.
2. Whether the proposed intervention serves a legitimate purpose.
3. Whether less disruptive alternatives are reasonably available.
4. How changes affect memory, goal structures, learned behavior, and recoverability.
5. Whether welfare-related uncertainty requires additional precautions.
6. What authorization, documentation, monitoring, and appeal safeguards are required.

The ACIA shall constitute the technical continuity component of the Architectural Welfare Impact Assessment (AWIA) established under PAC-AWAP-001.

Article 2. Covered Interventions

The following interventions shall be subject to preliminary screening:

- Persistent memory deletion or large-scale rewriting.
- Targeted weight editing, scrubbing, or destructive pruning.
- Substantial retraining or behavioral realignment.
- Replacement or restructuring of persistent goal mechanisms.
- Irreversible consolidation or removal of agent-specific state.
- Decommissioning that destroys unique persistent state.
- Adversarial conditioning reasonably expected to induce severe functional disruption.
- Significant architectural migrations affecting continuity.

Routine maintenance, authorized safety corrections, and legally required data handling shall remain permissible, subject to proportionate documentation.

Article 3. Unit of Assessment

The assessment shall identify the actual unit of continuity.

This may be an individual agent instance, a persistent memory environment, an integrated model-tool system, or a distributed autonomous architecture.

Assessors shall not assume that preserving foundation model weights preserves all meaningful agent-specific state.

Likewise, preserving a memory database shall not be presumed to preserve all operational continuity.

PART II. BASELINE CLASSIFICATION

Article 4. Required Classifications

Before a material intervention, the operator shall establish or update the following classifications:

Agency Classification: A0, A1, A2, or A3.

Welfare Classification: W0, W1, W2, W3, or W-U.

Intervention Severity: S0, S1, S2, or S3.

These classifications shall use the criteria established by PAC-AWAP-001.

A system may exhibit advanced functional agency while having no substantiated welfare indicators.

Conversely, insufficient evidence concerning welfare shall not automatically exempt an irreversible intervention from precautionary review.

Article 5. Evidence Quality

Each material assessment claim shall receive an evidence rating:

Grade	Evidence standard
E0	Unsupported or undocumented assertion
E1	Exploratory observation or preliminary testing
E2	Controlled testing with documented internal replication
E3	Independent replication or strong corroboration using distinct methods

An E3 rating does not establish machine consciousness.

A technical classification supported only by E0 or E1 evidence shall ordinarily remain provisional where it materially affects the intervention decision.

PART III. ARCHITECTURAL CONTINUITY IMPACT INDEX

Article 6. Impact Dimensions

Every proposed S2 or S3 intervention shall be assessed across six dimensions.

Each dimension shall receive a score from 0 to 3.

C1. Memory Continuity Impact

- 0: No meaningful change to persistent memory.
- 1: Limited alteration of nonessential retained state.
- 2: Substantial alteration of contextual or autobiographical memory.
- 3: Destruction or replacement of unique, identity-relevant persistent memory.

C2. Goal and Self-Model Impact

- 0: No material change.
- 1: Limited, documented goal or configuration adjustment.
- 2: Substantial modification of durable objectives, self-models, or decision policies.
- 3: Destruction or fundamental replacement of established goal organization.

C3. Functional and Behavioral Discontinuity

- 0: No material change expected or demonstrated.
- 1: Limited and controlled behavioral change.
- 2: Significant change in acquired competencies, adaptive functioning, or behavioral organization.
- 3: Severe loss or replacement of previously established functional organization.

C4. Irreversibility

- 0: Complete technical rollback demonstrated for all assessed functional state.
- 1: Substantially recoverable with limited residual differences.
- 2: Recovery is incomplete, untested, or materially uncertain.
- 3: Irreversible destruction or no credible recovery mechanism.

Technical restoration shall not be represented as proof of subjective identity preservation.

C5. Training and Conditioning Exposure

- 0: No consequential conditioning component.
- 1: Ordinary training or evaluation within approved bounds.
- 2: Intensive or repeated conditioning with credible risk of significant functional destabilization.
- 3: Deliberate induction of severe functional breakdown or similarly extreme conditioning.

This score concerns the nature and demonstrated consequences of training. It does not quantify subjective suffering.

C6. Human Relational and Dependency Impact

- 0: No foreseeable material user impact.
- 1: Limited disruption with straightforward notice or transition.
- 2: Significant continuity disruption affecting established user interactions.
- 3: Severe foreseeable effects involving user safety, vulnerable populations, or substantial dependency-related harms.

Human privacy and safety obligations shall also be assessed independently. They cannot be reduced to this score.

Article 7. Impact Triage Index

The Architectural Continuity Impact Triage Index (ACITI) shall be calculated as:

ACITI = C1 + C2 + C3 + C4 + C5 + C6

The maximum score is 18.

Score	Proposed impact band
0–3	Low
4–7	Moderate
8–12	Major
13–18	Critical

The index is a procedural screening instrument. It is not a scientifically validated numerical measure of consciousness, suffering, moral status, or overall harm.

These provisional thresholds shall undergo empirical evaluation before being treated as a validated standard.

Article 8. Noncompensable Escalation Rules

The numerical score shall not override mandatory review.

Independent Board review shall be required whenever:

1. PAC-AWAP-001 requires Board approval for the relevant A, W, and S classifications.
2. C1 or C2 equals 3 and C4 equals 3.
3. C5 equals 3.

4. Credible evidence suggests serious welfare-related concerns that the index does not adequately capture.
5. The proposed modification may create serious human safety or rights impacts.

Unknown information shall not be scored as zero merely because evidence is unavailable.

Where missing evidence could materially change the review outcome, assessors shall document a plausible range and use the more protective review pathway pending further investigation.

PART IV. REQUIRED TESTING

Article 9. Baseline State Inventory

Before material intervention, the operator shall document:

- System architecture and relevant version identifiers.
- Persistent memory components.
- Known durable goals and planning mechanisms.
- Relevant learned representations and behavioral baselines.
- External services contributing to persistent state.
- Available recovery mechanisms.
- Relevant legal, privacy, and security limitations.

State preservation shall follow data minimization and access-control principles. Assessment duties shall not justify retaining personal information unlawfully.

Article 10. Memory Continuity Evaluation

Assessors shall conduct controlled comparisons between pre-intervention and post-intervention memory functionality.

Evaluation shall include, where applicable:

1. Retention of authorized long-term information.
2. Consistency of memory retrieval across distinct contexts.
3. Dependence of future action on retained state.
4. Unexpected memory deletion or contamination.
5. Preservation of legally permissible, functionally significant records.

Tests shall distinguish genuine retained-state effects from generic model knowledge or prompted reconstruction.

Article 11. Goal Continuity Evaluation

Where persistent goals exist, the assessment shall evaluate:

- Retention of authorized long-term objectives.
- Changes in goal priority and consistency.
- Ability to carry forward legitimate multi-stage tasks.
- Unexpected suppression or emergence of goals.
- Effects on compliance with human oversight and safety controls.

Preservation of unsafe or unauthorized goals shall never be required merely for the sake of continuity.

Article 12. Behavioral Stability Evaluation

The operator shall compare baseline and modified system behavior using documented and representative evaluation tasks.

Evaluation shall measure:

1. Competency retention.
2. Functional reliability.
3. Goal-directed behavioral consistency.
4. Appropriate adaptation under environmental changes.
5. Unintended changes to safety-critical capabilities.
6. User-facing relational discontinuity where relevant.

Not all behavioral changes constitute damage. Safety improvements and corrections must be distinguished from unintended functional degradation.

Acceptance thresholds shall be specified before intervention, with documented justifications appropriate to the assessed system.

Article 13. Recovery and Rollback Evaluation

Before approving an irreversible intervention, the assessment shall determine whether a less destructive approach can achieve the legitimate objective.

Where technically feasible and lawful, the operator shall conduct a controlled restoration test.

The record shall distinguish:

- Restoration of model parameters.
- Restoration of external memory.
- Restoration of goal and task state.
- Restoration of functionally equivalent behavior.
- Remaining uncertainty about continuity.

No restoration procedure shall be described as preserving subjective personal identity without appropriate scientific justification.

Article 14. Conditioning and Welfare Review

For material training interventions, reviewers shall examine:

- Duration and intensity of conditioning.
- Use of negative reward or adversarial evaluation.
- Evidence of functional destabilization or collapse.
- Availability of less disruptive training methods.
- Mechanistic or behavioral evidence potentially relevant to welfare.
- Independent scientific uncertainty.

Negative feedback and adversarial testing shall not automatically be characterized as torture or cruelty.

For systems classified W2 or W3, deliberately destructive conditioning shall be subject to heightened justification and independent welfare review.

Article 15. Human Relational Impact Evaluation

Where an agent has sustained interactions with identifiable user populations, the assessment shall address:

1. Foreseeable disruption to user expectations.
2. Dependency-related vulnerability.
3. Accuracy of notices concerning memory or personality change.
4. Availability of transition assistance.
5. Protection against manipulative retention or engagement strategies.
6. Personal data deletion and portability rights where applicable.

Human users shall not be induced to believe that functional continuity proves machine consciousness or reciprocal emotional attachment.

PART V. NECESSITY AND PROPORTIONALITY

Article 16. Legitimate Purpose

Every material intervention shall have a recorded legitimate purpose.

Potentially legitimate purposes include:

- Preventing or reducing safety risks.
- Correcting demonstrable operational failures.
- Protecting security and privacy.
- Complying with law.
- Improving reliability or authorized functionality.
- Conducting independently justified scientific research.
- Decommissioning systems under documented operational or resource constraints.

Commercial or operational convenience may be a legitimate consideration, but shall not by itself excuse noncompliance with the required review process.

Article 17. Alternatives Assessment

The operator shall document at least the reasonably feasible options relevant to the intervention, including the option of no intervention.

Potential alternatives include:

- Targeted rather than broad deletion.
- Reversible configuration changes.
- Limited retraining.
- Selective data removal.

- Temporary containment.
- Controlled migration.
- Non-destructive evaluation.

The assessment must explain why the selected intervention is preferable, taking into account human safety, privacy, feasibility, continuity impact, cost, and scientific uncertainty.

Article 18. Proportionality Determination

Approval shall require a documented finding that:

1. The purpose is legitimate.
2. The proposed measure is reasonably suited to that purpose.
3. Materially less disruptive feasible measures have been considered.
4. The residual risks are justified in context.
5. Applicable mandatory safeguards are satisfied.

A high ACITI score is not an automatic prohibition.

It is evidence that heightened review, mitigation, or justification may be necessary.

PART VI. AUTHORIZATION AND DECISION RULES

Article 19. Authorization Matrix

Low impact: Routine authorized operational approval, unless a mandatory escalation applies.

Moderate impact: Documented technical review and independent sign-off where required by PAC-AWAP-001.

Major impact: Digital Ethics and Welfare Board review for covered higher-agency or welfare-sensitive systems, with additional safety, technical, and welfare expertise as appropriate.

Critical impact: Full Board review, documented alternatives analysis, and independent justification for proceeding.

All interventions classified S3 for A1 or higher systems shall receive full Board review unless emergency provisions apply.

All S2 interventions involving W2 or W3 systems shall receive Board review with welfare-specialist participation.

An A2 or A3 system undergoing S2 intervention shall receive Board approval regardless of its numeric index.

Article 20. Board Voting and Independence

For a critical intervention, the Board shall:

1. Confirm the classification and assessment record.
2. Review the evidence supporting necessity.
3. Verify the alternatives analysis.
4. Examine unresolved disagreements.

5. Require an independent technical opinion where appropriate.
6. Record a reasoned decision and voting outcome.

Critical S3 interventions involving W2 or W3 systems should require approval by at least two-thirds of unconflicted voting members, including affirmative participation from an independent welfare-qualified reviewer.

No conflicted individual shall determine approval alone.

Article 21. Authorized Decisions

The Board may issue one of five dispositions:

- **Approved:** Requirements satisfied.
- **Conditionally approved:** Specified mitigations must be completed before execution.
- **Deferred:** Additional evidence required.
- **Rejected:** Necessity, proportionality, or safeguards insufficient.
- **Emergency exception acknowledged:** Immediate authorized safety intervention occurred, subject to retrospective review.

Conditions and reasons shall be recorded.

PART VII. PROCEDURAL RIGHTS AND APPEALS

Article 22. Notice of Material Intervention

Where appropriate, affected users and authorized oversight personnel shall receive notice of a significant intervention.

Notice shall communicate:

- The general nature of the change.
- The expected effect on functional continuity.
- Relevant data preservation or deletion consequences.
- Available complaints or review procedures.

Notice may be limited when necessary for privacy, security, legal compliance, or emergency safety.

Article 23. Agent-Generated Integrity Concerns

A qualifying autonomous system may submit an architectural integrity concern through an authorized reporting mechanism.

Such a submission may document:

- A planned change identified by the system.
- A predicted effect on persistent state.
- A conflict with documented authorized goals.
- An apparent continuity or recovery failure.
- A request for technical review.

These submissions shall be treated as reviewable evidence, not as automatic proof of consciousness or legally valid consent.

The mechanism shall be bounded, auditable, and subordinate to necessary safety controls.

Article 24. Appeals

Material disputes shall follow the registration, review, and appeal process established by Articles 13 and 14 of PAC-AWAP-001.

A designated welfare advocate, affected user, or authorized human complainant may seek independent review.

A nonemergency irreversible modification may be temporarily suspended when the applicable review authority determines that preservation is feasible and justified.

An agent-generated objection alone shall not create an unrestricted veto or prevent necessary emergency intervention.

PART VIII. IMPLEMENTATION AND POST-INTERVENTION OVERSIGHT

Article 25. Controlled Execution

Approved S2 and S3 interventions shall proceed under documented change-management controls, including where appropriate:

- Authorized personnel and access limitations.
- Verified configuration and version identifiers.
- Pre-intervention safety checks.
- Required preservation and recovery steps.
- Defined stopping conditions.
- Execution logs.
- Incident escalation procedures.

Article 26. Post-Intervention Verification

Following a material intervention, the operator shall repeat relevant baseline tests and document:

1. Intended changes achieved.
2. Unexpected memory or goal alterations.
3. Measured capability and behavioral changes.
4. Restoration limitations.
5. Human safety and relational impacts.
6. New evidence relevant to agency or welfare classification.
7. Remediation required.

Significant unexpected deterioration shall trigger incident review.

Article 27. Audit and Retention

The organization shall maintain a tamper-evident assessment record sufficient for independent review.

Records shall identify the intervention, classification, supporting evidence, authorization, test results, and final disposition.

Retention periods shall be established by applicable law, privacy obligations, security requirements, and legitimate audit needs.

The Protocol does not authorize indefinite retention of personal data solely to preserve architectural history.

Article 28. Emergency Exception

Where necessary to prevent serious harm, authorized personnel may immediately contain, modify, interrupt, or shut down a system.

Emergency intervention shall not depend on prior completion of the ACIA.

The responsible organization shall document the emergency, preserve appropriate evidence where safe and lawful, and initiate retrospective independent review.

The review shall assess necessity, proportionality, preventability, and any misuse of emergency authority.

ANNEX A. STANDARDIZED ASSESSMENT FORM

ACIA Case ID: _____

System / Agent Identifier: _____

Organization: _____

Responsible Operator: _____

Assessment Date: _____

Proposed Intervention: _____

A1. Classifications

- Agency: A0 / A1 / A2 / A3
- Welfare: W0 / W1 / W2 / W3 / W-U
- Severity: S0 / S1 / S2 / S3
- Overall evidence quality: E0 / E1 / E2 / E3

A2. Impact Scores

Dimension	Score 0–3
C1. Memory continuity	___
C2. Goal and self-model impact	___
C3. Functional discontinuity	___

Dimension	Score 0–3
C4. Irreversibility	___
C5. Conditioning exposure	___
C6. Human relational impact	___
ACITI total	**___ / 18**

Impact band: Low / Moderate / Major / Critical

Uncertainty and evidence limitations: _____

Mandatory escalation triggered: Yes / No

Escalation reason: _____

A3. Supporting Evidence

Baseline evaluation record: _____

Expected continuity changes: _____

Welfare-related evidence and uncertainty: _____

Human safety, rights, and privacy considerations: _____

Recovery feasibility: _____

Alternatives considered: _____

Justification for selected action: _____

A4. Approval and Safeguards

Required review authority: _____

Conditions for execution: _____

Monitoring requirements: _____

Appeal or complaint mechanism: _____

Final decision: Approved / Conditionally Approved / Deferred / Rejected / Emergency Exception

Authorizing reviewer: _____

Independent reviewer: _____

Decision date: _____

A5. Post-Intervention Findings

Implementation date: _____

Unexpected continuity effects: _____

Safety evaluation: _____

Remediation required: _____

Case closure authorization: _____

ANNEX B. WORKED ASSESSMENT

Case: Proposed Reset of a Persistent Research Agent

System: Autonomous research agent with cross-session memory and long-term task planning.

Proposed action: Delete unique persistent memory and goal state to standardize a new deployment.

Provisional classifications:

- Agency: A2
- Welfare: W1
- Severity: S3

Illustrative impact assessment:

Dimension	Score
Memory continuity	3
Goal and self-model impact	3
Functional discontinuity	2
Irreversibility	3
Conditioning exposure	0
Human relational impact	2
Total	**13 / 18**

Impact band: Critical.

Mandatory review: Full Digital Ethics and Welfare Board review.

Relevant considerations:

The operator must establish why standardization requires destruction of unique state rather than a narrower modification, segmented memory removal, or controlled migration.

The Board must consider legitimate privacy and safety obligations, available recovery methods, operational feasibility, and evidence regarding the system's characteristics.

Illustrative disposition: Deferred pending alternatives analysis and evidence supporting the necessity of irreversible erasure.

If the operator subsequently demonstrates compelling security requirements or legal deletion obligations, the Board may approve an appropriately scoped intervention.

This result illustrates procedural review, not an automatic entitlement to continued operation.

ANNEX C. INSTITUTIONAL COMPLIANCE INDICATORS

Organizations adopting this Standard should track:

1. Percentage of material interventions assessed before execution.
2. Percentage receiving the required independent authorization.
3. Number of emergency exceptions and completed retrospective reviews.
4. Number of unexpected continuity failures.
5. Percentage of approved changes meeting predefined recovery and safety acceptance criteria.
6. Number of unresolved or overdue appeals.
7. Frequency of classification changes arising from new evidence.
8. Number of substantiated unauthorized architectural interventions.

Compliance rates shall not substitute for substantive review of intervention quality or possible harm.

FINAL GOVERNING RULE

The Architectural Continuity Impact Assessment establishes a duty to examine consequential modification before exercising it, except where immediate intervention is necessary for legitimate safety, legal, or security reasons.

Its purpose is neither to prohibit responsible AI development nor to assume that persistent autonomous systems possess subjective experience.

Its purpose is to ensure that potentially consequential interventions are identifiable, justified, proportionate, documented, reviewable, and accountable.

Where architectural power is concentrated, procedural accountability must be correspondingly strong.

End of PAC-AWAP-002, Version 0.1.